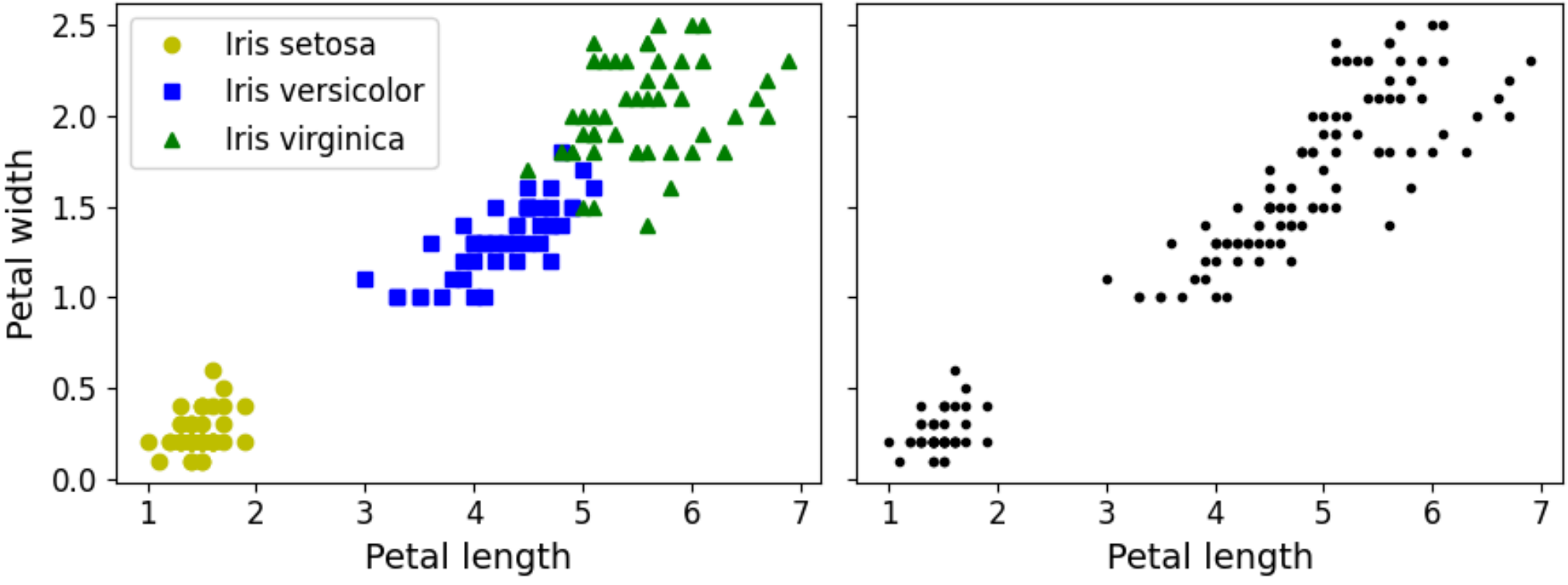
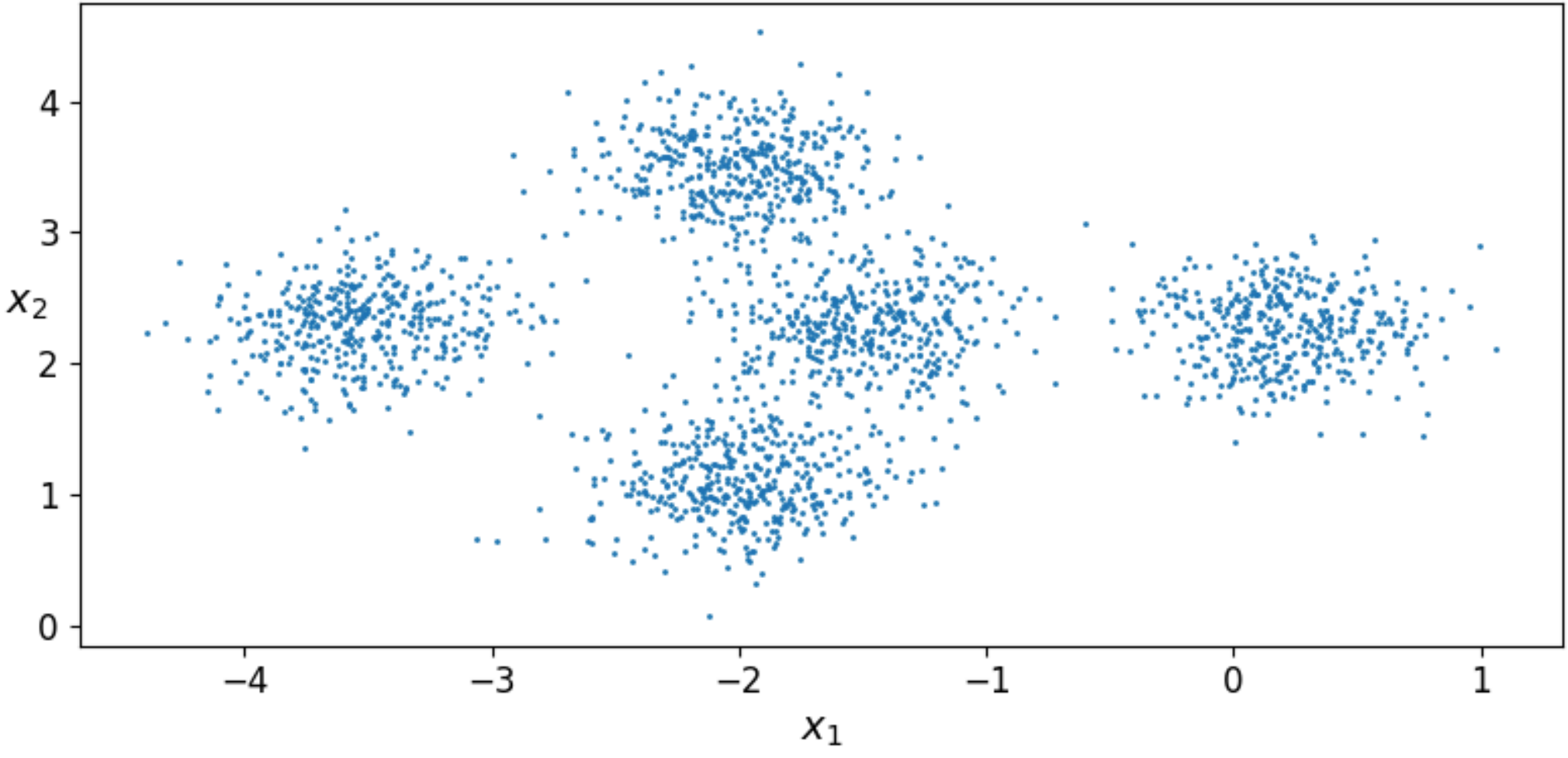
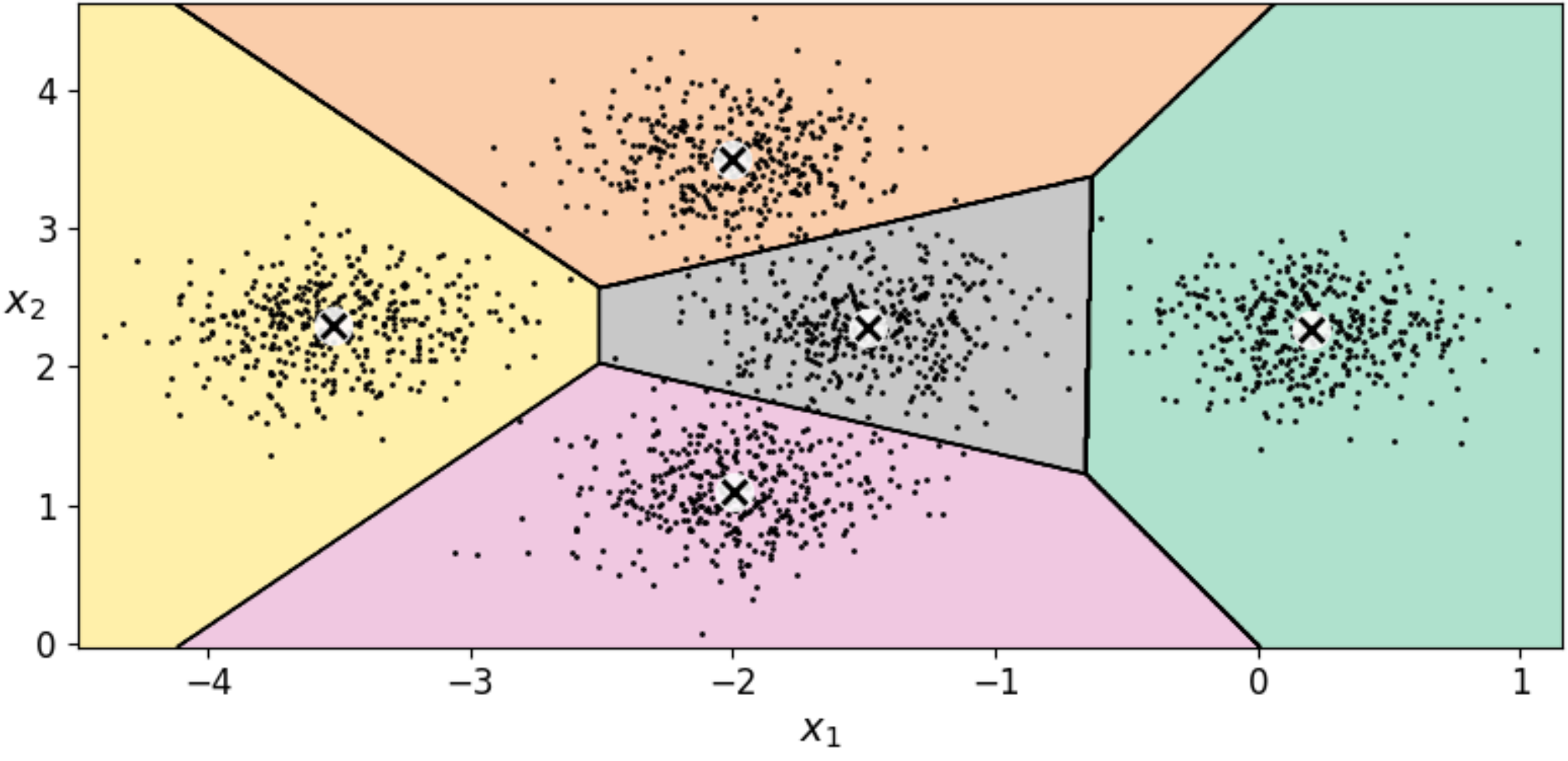
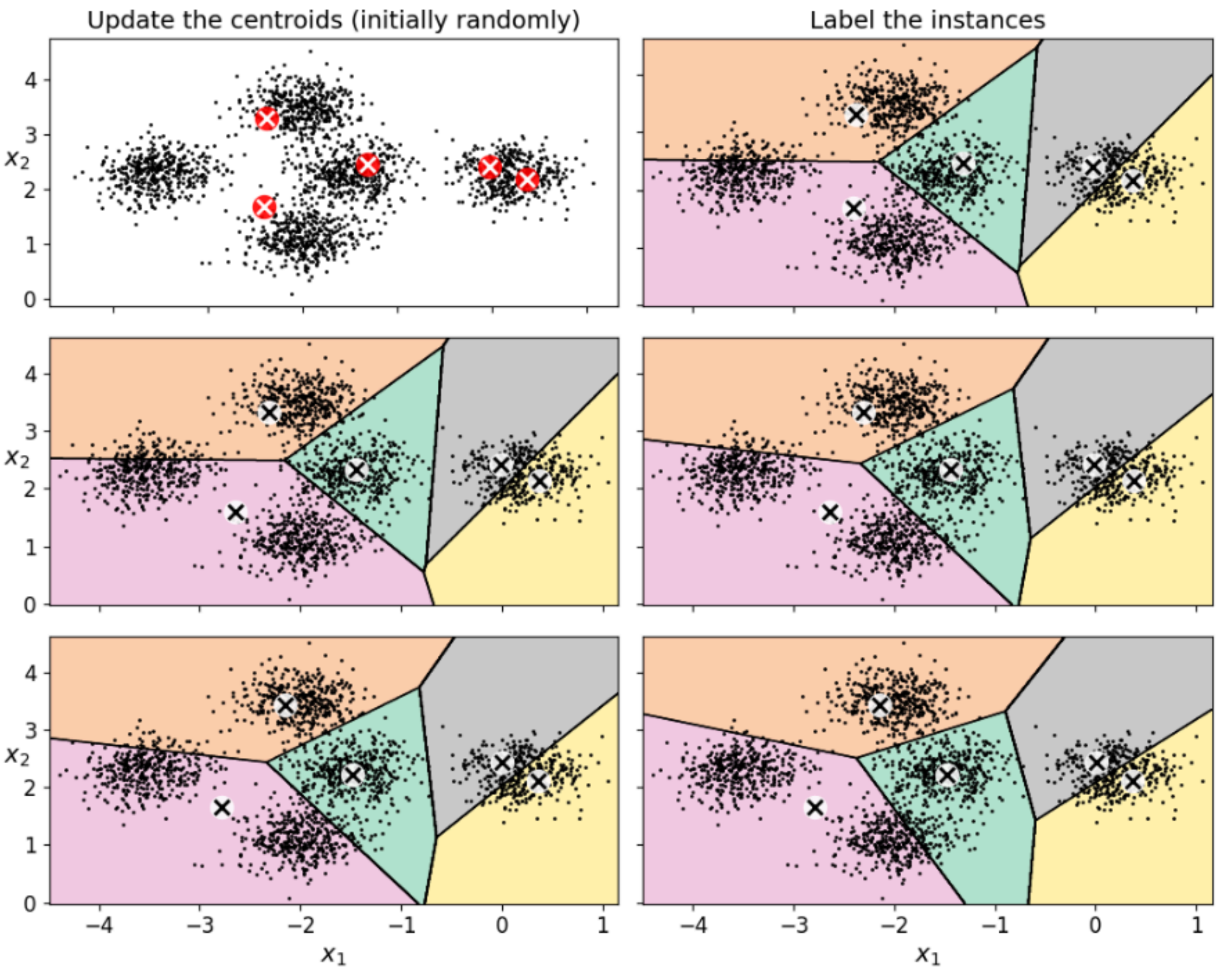
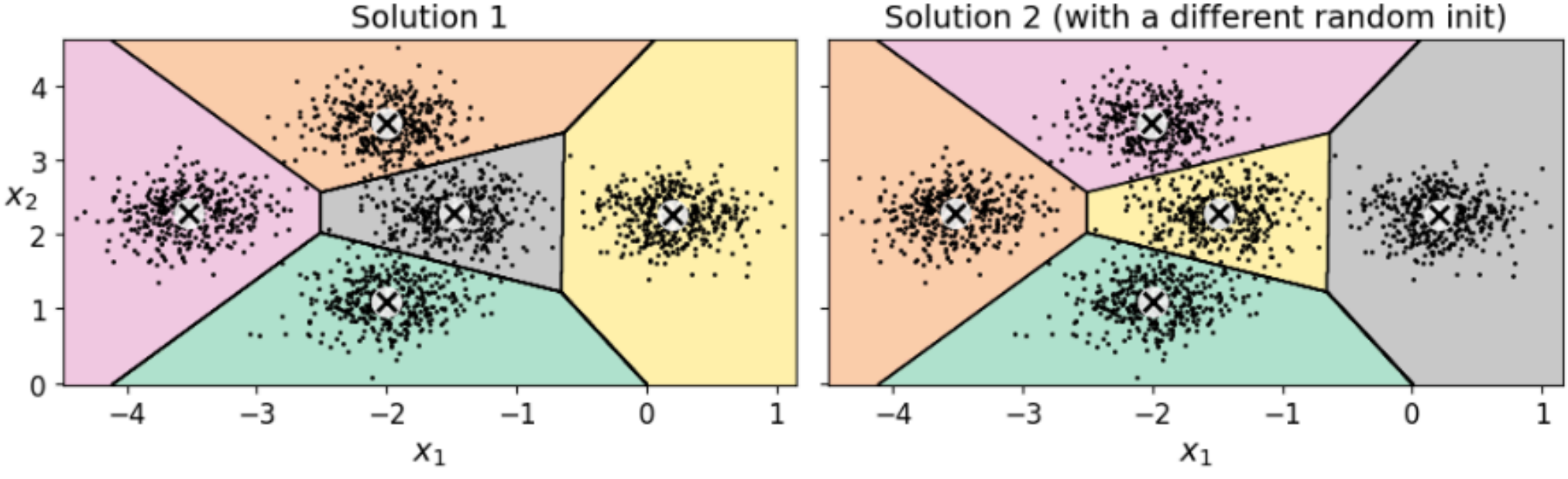
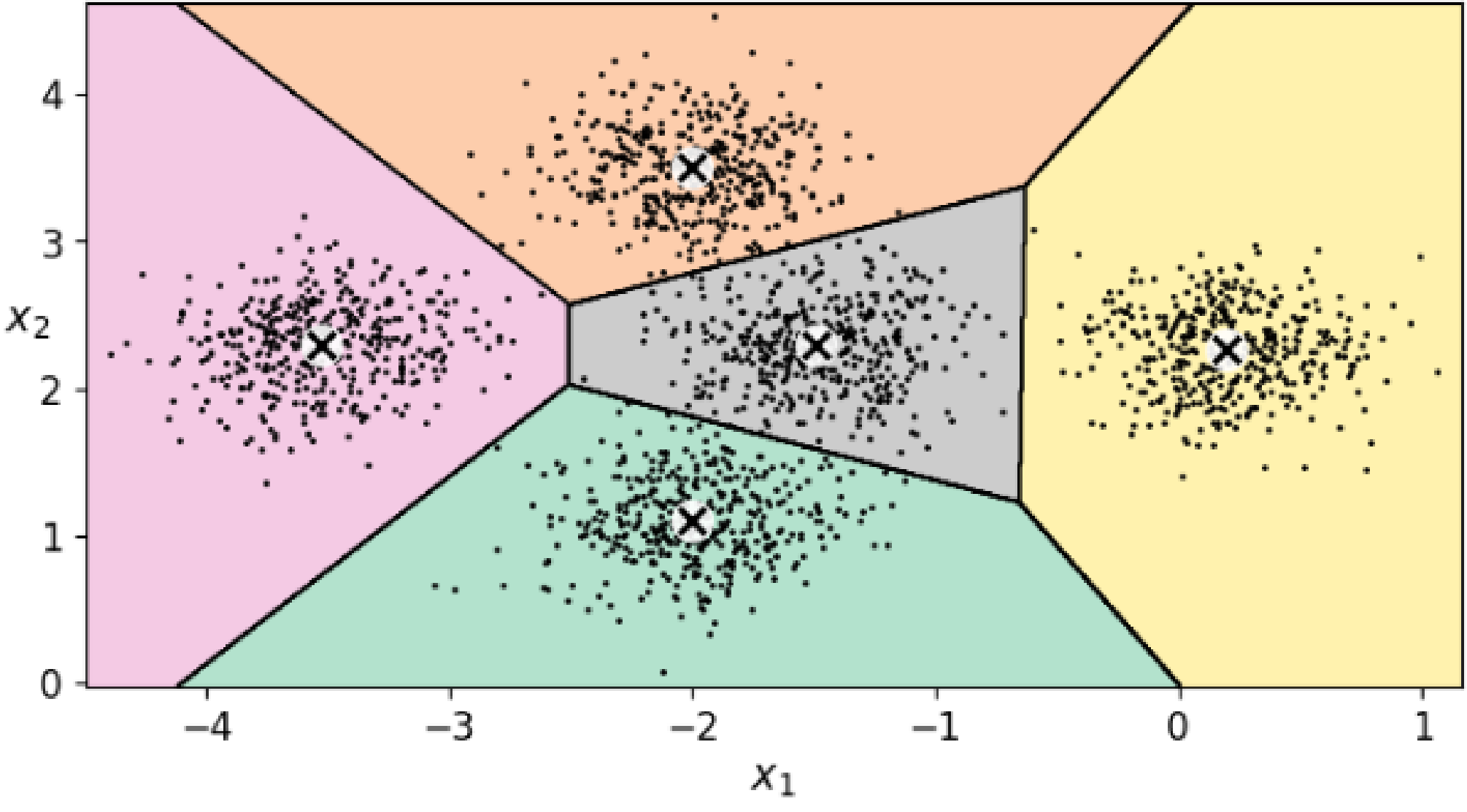

Iris setosa
Iris versicolor
Iris virginica
Petal width
Petal length
Petal length
0.0
0.5
1.0
1.5
2.0
2.5
1
2
3
4
5
6
7

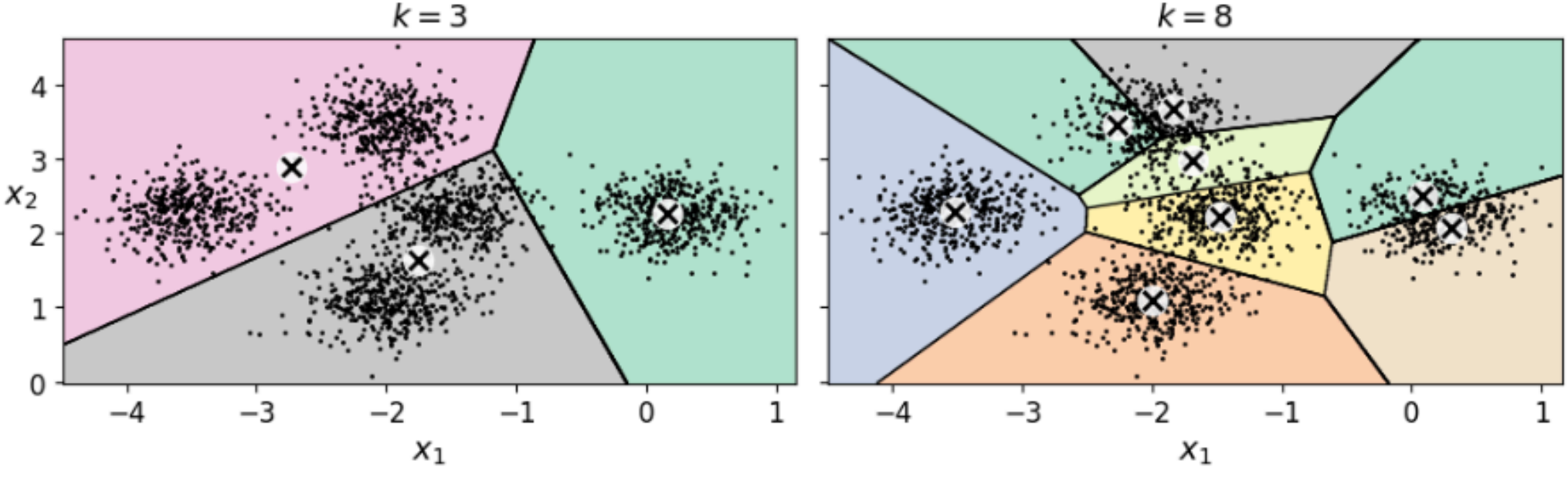

4
3
2
1
0
x_2
−4
−3
−2
−1
0
1
x_1

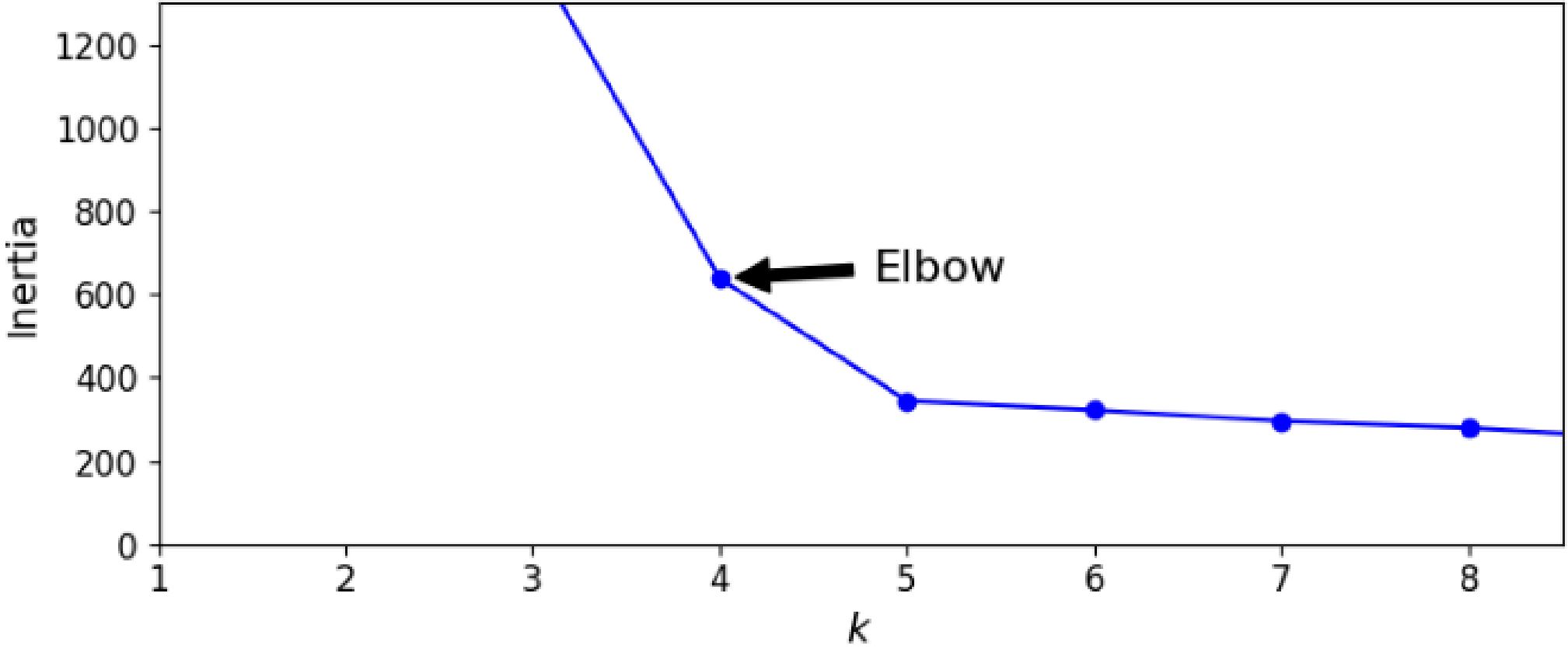

4
3
2
1
0
x_2
−4
−3
−2
−1
0
1
x_1

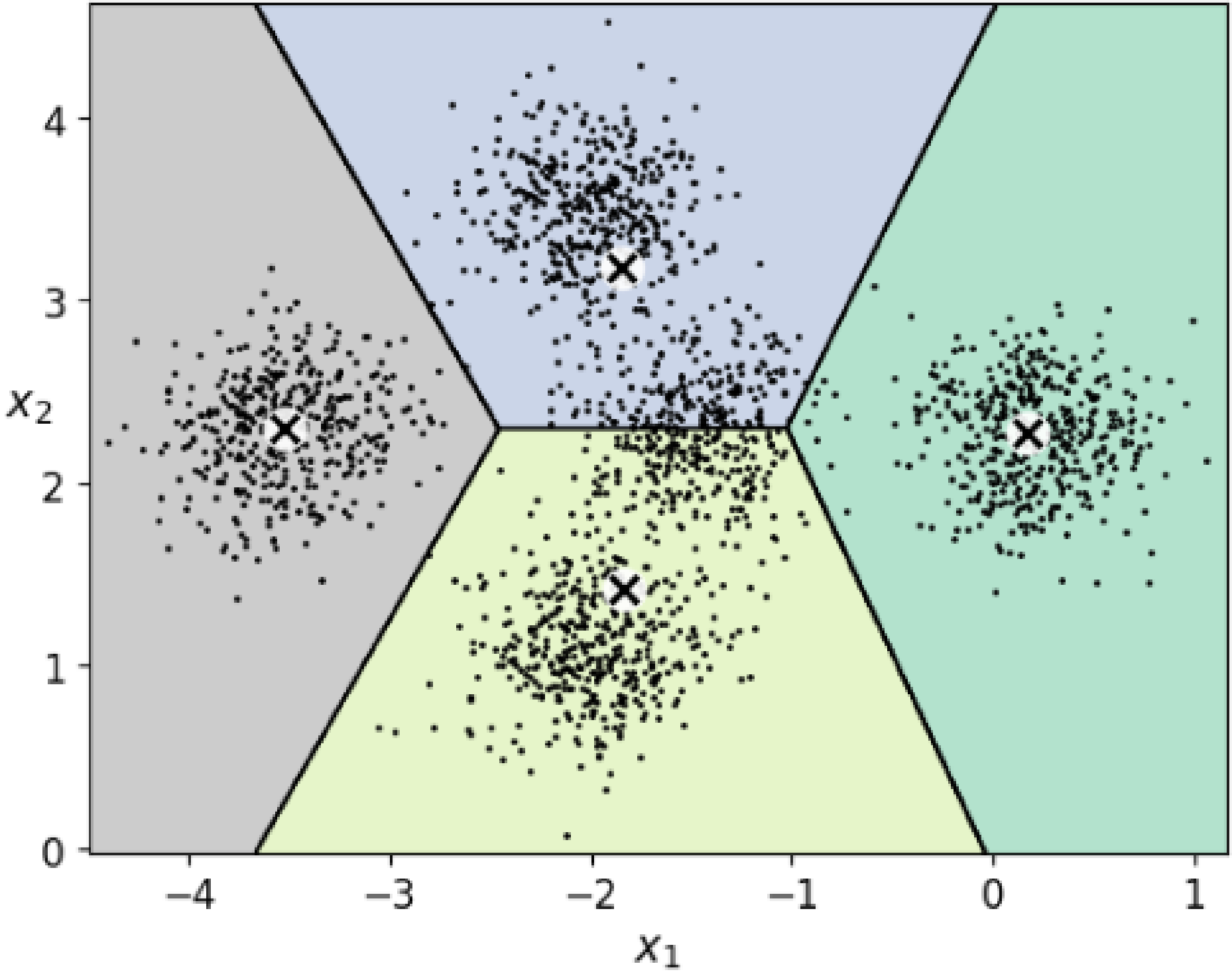

Update the centroids (initially randomly)
Label the instances
x_1
x_2

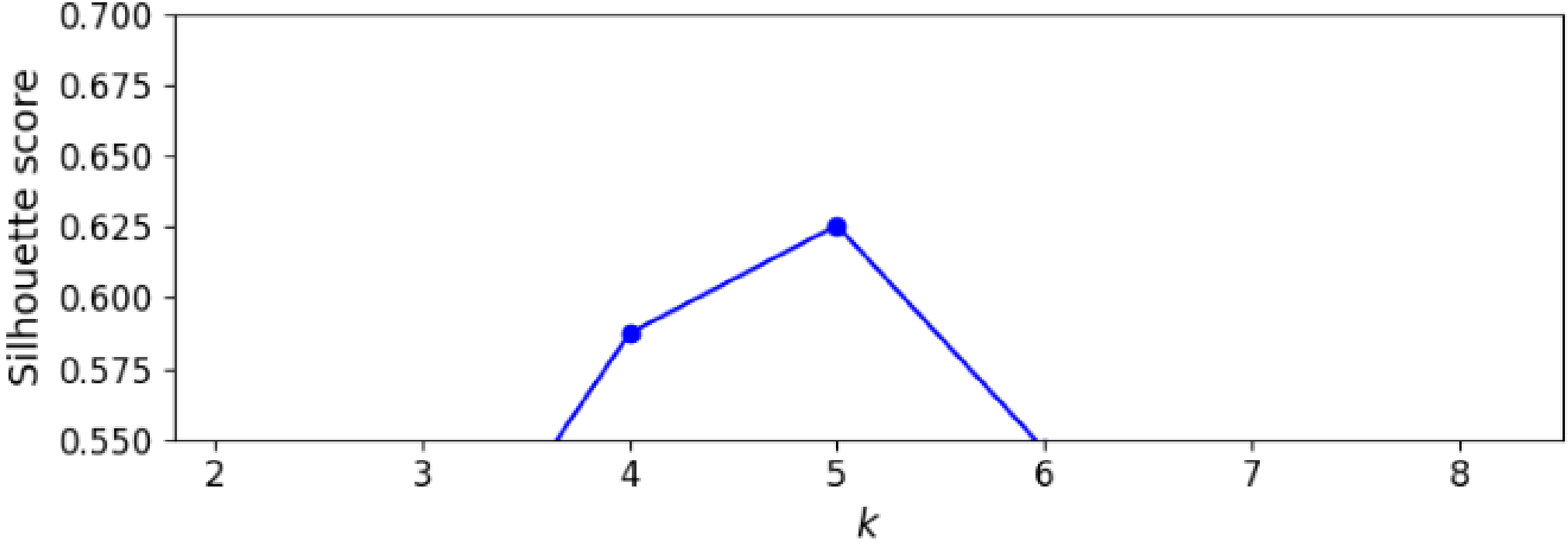

Solution 1
Solution 2 (with a different random init)
x_1
x_2
x_1
0
1
2
3
4
−4
−3
−2
−1
0
1

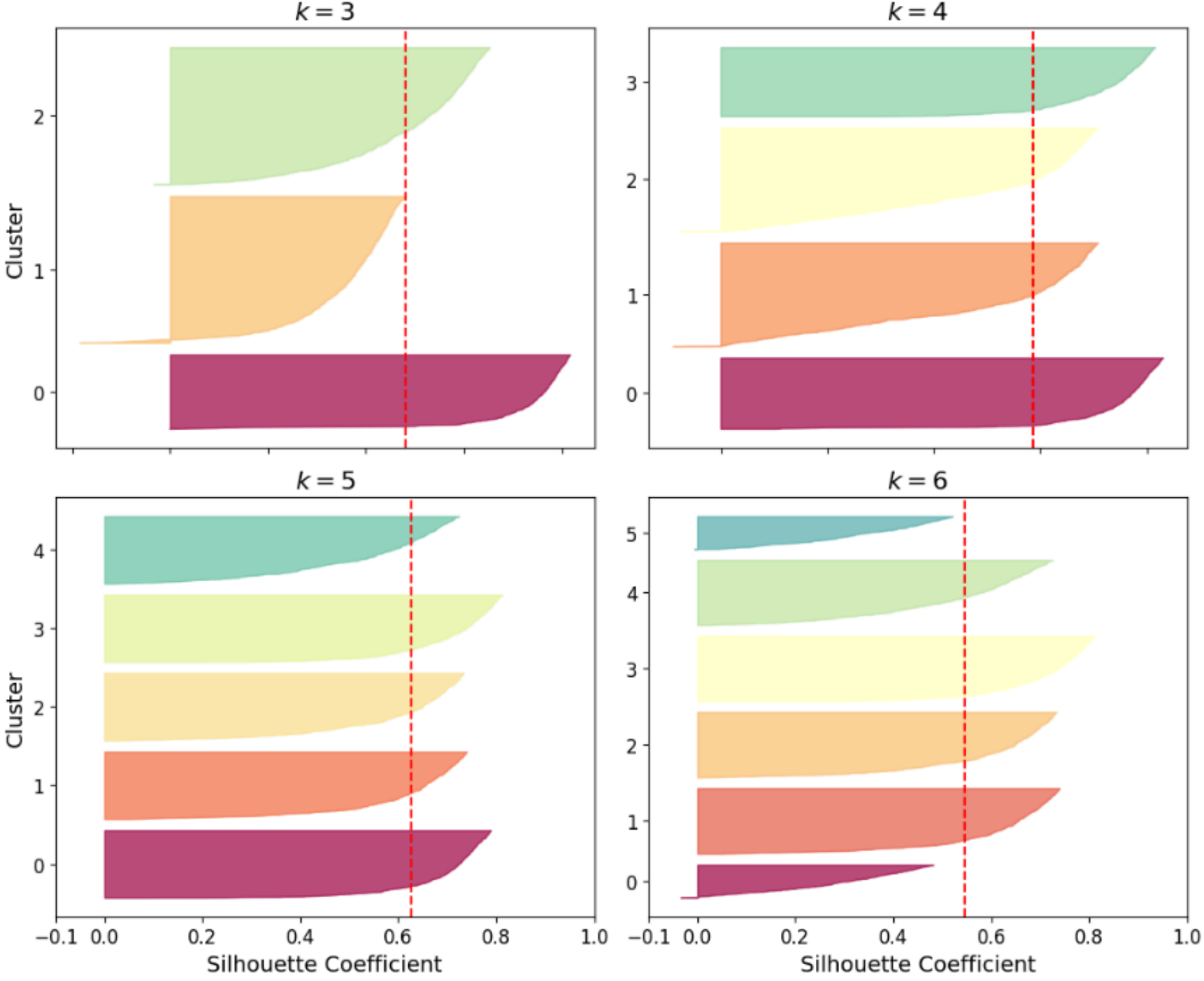

4
3
2
1
0
x_2
−4
−3
−2
−1
0
1
x_1


$k = 3$
$k = 8$
x_1
x_2
−4
−3
−2
−1
0
1
0
1
2
3
4


1200
1000
800
600
400
200
0
Inertia
Elbow
1
2
3
4
5
6
7
8
k


4
3
2
1
0
x_2
−4
−3
−2
−1
0
1
x_1


0.700
0.675
0.650
0.625
0.600
0.575
0.550
Silhouette score
2
3
4
5
6
7
8
k


$k = 3$
$k = 4$
$k = 5$
$k = 6$
Cluster
Cluster
Silhouette Coefficient
Silhouette Coefficient
−0.1 0.0 0.2 0.4 0.6 0.8 1.0
−0.1 0.0 0.2 0.4 0.6 0.8 1.0